



Moove Education Platform

Software Requirements Specification (SRS)

Version 3.1

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Contents

List of Tables	3
List of Figures	4
0.1 Document Management and Quality Assurance	5
0.1.1 Document Purpose and Scope	5
0.1.2 Document Management	5
0.1.3 Tools and Methodology	5
0.1.4 Compliance Framework	5
0.1.5 Quality Assurance Measures	6
0.2 Introduction	6
0.2.1 Purpose	6
0.2.2 Scope	6
0.2.3 System Overview and Actors	6
0.2.4 System Constraints	7
0.2.5 Assumptions and Dependencies	7
0.3 Functional Requirements – Educator Features	7
0.3.1 Course Creation & Planning Module	7
0.3.2 Instructional Design Concepts	10
0.3.3 Learner Engagement & Communication Module	10
0.3.4 Content Creation & Management Module	12
0.3.5 Components & Activities Module	14
0.3.6 Data & Analytics Module	17
0.3.7 Accessibility Module	20
0.3.8 Import/Export / OLX Module	21
0.4 Functional Requirements – Learner Features	22
0.4.1 Dashboard & Profile Module	22
0.4.2 Course Navigation Module	27
0.4.3 Enrollment Module	27
0.4.4 Course Prerequisites Module	28
0.4.5 Course Content Access Module	30
0.4.6 Course Progress Module	32
0.4.7 Certificates & Badges Module	34
0.4.8 Video Player Module	35
0.4.9 Discussions Module	36
0.4.10 Assignments & Assessments Module	38
0.4.11 Team Projects Module	43
0.4.12 Search Module	46
0.4.13 Bookmarking Module	48
0.4.14 Notifications & Preferences Module	49
0.4.15 Course Wiki Module	51

0.5	Security and Compliance Requirements (Non-Functional)	52
0.5.1	Access Control (CMCSRS 6.17)	52
0.5.2	Cryptography (CMCSRS 6.18)	52
0.5.3	Network Security (CMCSRS 5.7.c)	52
0.5.4	Logging and Monitoring (CMCSRS 6.25)	52
0.5.5	Incident Response and Business Continuity (CMCSRS 6.26, 6.19) .	53
0.5.6	Personnel Security (CMCSRS 5.6, 6.37)	53
0.5.7	Compliance and Audit (CMCSRS 6.21, 6.22, 6.23)	53
0.6	Non-Functional Requirements	53
0.6.1	Performance	53
0.6.2	Security	53
0.6.3	Availability	54
0.6.4	Maintainability	54
0.6.5	Scalability	54
0.6.6	Usability	54
0.7	Traceability Matrix	54
0.8	Compliance Matrix	55
0.9	Glossary	56
0.10	References	56
.1	Full Apache License 2.0 Text	58

List of Tables

List of Figures

0.1 Document Management and Quality Assurance

0.1.1 Document Purpose and Scope

This Software Requirements Specification (SRS) serves as the authoritative source of system requirements for the Moove Education Platform—an Open edX-based learning management system deployed using Tutor. This document is a living artifact maintained in a Git repository, subject to version control, peer review, and continuous improvement.

0.1.2 Document Management

- **Version Control:** All documents are managed in a Git repository (`git@github.com:Moove-Digital-Solutions/moove-openedx-docs.git`)
- **Branching Strategy:** Main branch contains production-ready documentation; feature branches are used for improvements
- **Review Process:** All changes require peer review before merging
- **Release Process:** Documentation versions align with software releases

0.1.3 Tools and Methodology

- **Documentation Tool:** LaTeX with a custom preamble for consistent styling
- **Diagramming:** PlantUML for UML diagrams, with PNG exports for inclusion
- **Screenshots:** Captured from the production environment
- **Build Automation:** `build.sh` script compiles all documents with `latexmk`
- **Continuous Improvement:** Regular reviews against INSA regulations

0.1.4 Compliance Framework

This document and the system it describes are designed to comply with:

- Critical Mass Cyber Security Requirement Standard (CMCSRS) v2.0
- FDRE National Cybersecurity Policy (2024)
- Computer Crime Proclamation No. 958/2016
- Electronic Signature Proclamation No. 1072/2018
- Cyber Security Audit and Evaluation Guideline v1.0
- Cyber Security Risk Assessment Framework v1.0

0.1.5 Quality Assurance Measures

- **Automated Validation:** The `assess_security.sh` script validates security controls
- **Vulnerability Scanning:** Trivy performs daily container image scanning
- **Log Review:** Centralized logging via Tutor log analysis
- **Health Checks:** Container health checks configured for all services

0.2 Introduction

0.2.1 Purpose

This document defines the functional and non-functional requirements for the Moove Education Platform, an Open edX-based learning management system deployed using Tutor. It serves as the living source of truth for system capabilities, guiding continuous development, internal quality assurance, and security assessment activities. The requirements are aligned with Ethiopian regulatory frameworks, including the Critical Mass Cyber Security Requirement Standard (CMCSRS), the National Cybersecurity Policy, the Computer Crime Proclamation, and the Electronic Signature Proclamation.

The SRS has been developed following IEEE Std 830-1998 guidelines for software requirements specifications, ensuring completeness, consistency, and traceability.

0.2.2 Scope

The system provides a comprehensive online learning environment supporting:

- Course creation, authoring, and management (Educator features).
- Learner enrollment, progression, and engagement (Learner features).
- Content delivery (video, text, interactive components).
- Assessments (quizzes, assignments, peer reviews, proctored exams).
- Collaboration (discussions, teams, wikis).
- Analytics and reporting.
- Administrative controls.

0.2.3 System Overview and Actors

The platform is a client-server application based on Open edX, deployed in Docker containers orchestrated by Tutor. It uses a microservices architecture with separate containers for LMS, CMS, MFE, databases, and supporting services.

Actors

- **Educator (Instructor/Staff):** Creates and manages courses, creates content, assesses learners, moderates discussions, views analytics.
- **Learner (Student):** Enrolls in courses, accesses content, completes assessments, participates in discussions, tracks progress.
- **Administrator:** Manages users, courses, site configuration, and system health.
- **System:** Automated processes (notifications, email, background tasks).

0.2.4 System Constraints

- The system must comply with Ethiopian laws and regulations, including the Computer Crime Proclamation and the Electronic Signature Proclamation.
- All user data must be stored and processed within the country's jurisdiction.
- The system must support English and Amharic (planned) interfaces.
- The system shall be accessible via modern web browsers and mobile devices.

0.2.5 Assumptions and Dependencies

- The platform assumes reliable internet connectivity for both learners and educators.
- Third-party services (e.g., video hosting, proctoring) are available and secured.
- The underlying Tutor deployment and container infrastructure are maintained and updated regularly.

0.3 Functional Requirements – Educator Features

0.3.1 Course Creation & Planning Module

General Description Provides tools for educators to create, plan, and structure courses.

Requirement UFR-OE-001-01: Create a New Course

1. **Title:** Create a new course.
2. **UML Traceability:** Use Case: “Create Course”; Activity: “Course Creation Flow”.
3. **Actor:** Educator (Instructor/Staff).
4. **Pre-conditions:**
 - Educator is logged in with course creation permissions.
 - The organisation code, course number, and run combination is unique.
5. **Post-conditions:**

- A new course skeleton is created in the CMS.
- The course appears in the educator's course list.
- An audit log entry is recorded.

6. Normal Flow:

- (a) Educator navigates to Studio (CMS).
- (b) Clicks "New Course".
- (c) Enters course title, organisation code, course number, and course run.
- (d) Sets start date, enrollment start/end dates, and course visibility.
- (e) Clicks "Create".
- (f) System validates uniqueness and creates the course ID.

7. Alternative Flows:

- Alt 1: Duplicate course ID → validation error.

8. Business Rules: Course ID format: org+course+run; must be unique.

9. Input/Output: Input: title, org, number, run, start date, enrollment dates. Output: Course object with ID.

10. Acceptance Criteria:

Given I am an educator
When I create a course with title "Introduction to Physics"
Then I receive a 201 Created
And the course appears in my course list

Requirement UFR-OE-001-02: Plan a New Course (Outline)

1. **Title:** Plan course outline.
2. **UML Traceability:** Activity: "Course Outline Flow".
3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Course outline (sections, subsections, units) is defined.

6. Normal Flow:

- (a) Educator opens the course outline.
- (b) Clicks "Add Section" and enters a name.
- (c) Within a section, clicks "Add Subsection" and enters a name.
- (d) Within a subsection, clicks "Add Unit" to create content pages.
- (e) Drags and drops to reorder.

7. Alternative Flows: None.

8. Business Rules: Sections, subsections, and units have a hierarchical order; re-ordering is allowed.

9. Input/Output: Input: section/subsection/unit metadata. Output: updated course structure.

10. Acceptance Criteria:

```
Given I am an educator with a course
When I add a section named "Week 1"
Then the section appears in the course outline
And I can add subsections under it
```

Requirement UFR-OE-001-03: Enroll Learners in a Course

1. Title: Enroll learners manually.

2. UML Traceability: Use Case: “Manage Enrollments”.

3. Actor: Educator.

4. Pre: Course exists and enrollment is open.

5. Post: Learner is enrolled.

6. Normal Flow:

- (a) Educator goes to “Instructor Dashboard” → “Enrollment”.
- (b) Enters learner email(s) and selects enrollment track (audit/verified).
- (c) Clicks “Enroll”.
- (d) System sends confirmation email.

7. Alternative Flows: Bulk enrollment via CSV upload.

8. Business Rules: CSV format: email,enrollment_mode; only valid emails are processed.

9. Input/Output: Input: email(s), enrollment mode. Output: enrollment confirmation.

10. Acceptance Criteria:

```
Given I am an educator
When I enroll a learner with email student@example.com in verified mode
Then the learner is enrolled and receives a confirmation email
```

0.3.2 Instructional Design Concepts

Requirement UFR-OE-002-01: Apply Instructional Design Principles

1. **Title:** Apply instructional design concepts.
2. **UML Traceability:** – (guideline).
3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Course is structured using sound pedagogical design.
6. **Normal Flow:**
 - (a) Educator defines learning objectives for each section.
 - (b) Creates assessments aligned with objectives.
 - (c) Uses Open edX best practices (backward design, Bloom’s taxonomy).
7. **Alternative Flows:** None.
8. **Business Rules:** None.
9. **Input/Output:** None; this is a guideline.
10. **Acceptance Criteria:** The course structure follows backward design and includes assessments at appropriate levels.

0.3.3 Learner Engagement & Communication Module

Requirement UFR-OE-003-01: Course Discussions

1. **Title:** Configure course discussions.
2. **UML Traceability:** Use Case: “Manage Discussions”.
3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Discussion forums enabled.
6. **Normal Flow:**
 - (a) Educator goes to “Advanced Settings” and enables discussions.
 - (b) Configures topics and moderation policies.
7. **Alternative Flows:** None.
8. **Business Rules:** Discussions are enabled only if the platform supports them.
9. **Input/Output:** Input: enable flag, topics. Output: updated course settings.
10. **Acceptance Criteria:**

```
Given I am an educator
When I enable discussions and set topics
Then the Discussions tab appears in the course navigation
```

Requirement UFR-OE-003-02: Bulk Emails

1. **Title:** Send bulk emails to learners.
2. **UML Traceability:** Sequence: “Bulk Email Flow”.
3. **Actor:** Educator.
4. **Pre:** Course has enrolled learners.
5. **Post:** Email sent asynchronously.
6. **Normal Flow:**
 - (a) Educator composes email, selects recipients (all learners, specific segments).
 - (b) Clicks “Send”.
 - (c) System queues via Celery.
7. **Alternative Flows:** None.
8. **Business Rules:** Emails are sent in batches; rate limits apply.
9. **Input/Output:** Input: subject, body, recipient list. Output: task ID.
10. **Acceptance Criteria:**

```
Given I am an educator
When I send a bulk email to all learners
Then the email is queued and delivered within 5 minutes
```

Requirement UFR-OE-003-03: Course Updates

1. **Title:** Publish course updates.
2. **UML Traceability:** – (simple notification).
3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Update appears on course home page.
6. **Normal Flow:** Educator creates an update post (title, content) and publishes.
7. **Alternative Flows:** None.
8. **Business Rules:** Updates are visible to all enrolled learners.

9. Input/Output: Input: title, content. Output: published update.

10. Acceptance Criteria:

```
Given I am an educator
When I publish a course update
Then it appears on the course home page immediately
```

0.3.4 Content Creation & Management Module

Requirement UFR-OE-004-01: Create and Manage Components

1. Title: Add and manage course components.

2. UML Traceability: Activity: “Component Management”.

3. Actor: Educator.

4. Pre: Unit exists.

5. Post: Component added to unit.

6. Normal Flow:

- (a) Educator selects a unit.
- (b) Clicks “Add New Component” and chooses type (HTML, Video, Problem, Discussion, etc.).
- (c) Configures component content.
- (d) Saves.

7. Alternative Flows: None.

8. Business Rules: Component types are defined by the XBlock system.

9. Input/Output: Input: component type, configuration. Output: component stored.

10. Acceptance Criteria:

```
Given I am an educator
When I add a Video component to a unit
Then the video appears in the unit content
```

Requirement UFR-OE-004-02: Manage Text Components

1. **Title:** Create and edit HTML/text components.
2. **Actor:** Educator.
3. **Pre:** Unit exists.
4. **Post:** Text component saved.
5. **Normal Flow:** Uses rich text editor or raw HTML; supports LaTeX, code snippets.
6. **Alternative Flows:** None.
7. **Business Rules:** HTML must be sanitized to prevent XSS.
8. **Input/Output:** Input: HTML content. Output: stored component.
9. **Acceptance Criteria:**

```
Given I am an educator
When I edit a text component with LaTeX
Then the rendered content displays the math correctly
```

Requirement UFR-OE-004-03: Manage Video Components

1. **Title:** Upload and manage video content.
2. **UML Traceability:** Use Case: “Manage Video”.
3. **Actor:** Educator.
4. **Pre:** Video file available.
5. **Post:** Video player embedded.
6. **Normal Flow:** Uploads video file or enters URL; system processes and displays player with transcripts, speed control.
7. **Alternative Flows:** URL from YouTube/Vimeo.
8. **Business Rules:** Video must be in supported formats; transcripts are optional.
9. **Input/Output:** Input: video file or URL, transcript (optional). Output: video player.
10. **Acceptance Criteria:**

```
Given I am an educator
When I upload a video file
Then the video player appears and plays the video
```

Requirement UFR-OE-004-04: Control Content Visibility & Access

1. **Title:** Control content visibility and access.
2. **UML Traceability:** – (settings).
3. **Actor:** Educator.
4. **Pre:** Content exists.
5. **Post:** Visibility rules applied.
6. **Normal Flow:** Sets release dates, prerequisites, and visibility rules.
7. **Alternative Flows:** None.
8. **Business Rules:** Release dates must be in the future; prerequisites must be valid content IDs.
9. **Input/Output:** Input: release date, prerequisites. Output: updated content visibility.
10. **Acceptance Criteria:**

```
Given I am an educator
When I set a release date in the future
Then the content is hidden until that date
```

0.3.5 Components & Activities Module**Requirement UFR-OE-005-01: Dropdown Problem**

1. **Title:** Create dropdown problem.
2. **UML Traceability:** – (component type).
3. **Actor:** Educator.
4. **Pre:** Unit exists.
5. **Post:** Problem added to unit.
6. **Normal Flow:**
 - (a) Adds a Problem component.
 - (b) Selects “Dropdown” type.
 - (c) Enters question and options.
 - (d) Marks correct answer.
 - (e) Saves.
7. **Alternative Flows:** None.

8. **Business Rules:** One correct answer.
9. **Input/Output:** Input: question, options, correct answer. Output: problem stored.
10. **Acceptance Criteria:**

Given I am an educator
When I create a dropdown problem
Then the problem is displayed and grades the learner's selection

Requirement UFR-OE-005-02: Multi-Select Problem

1. **Title:** Create multi-select problem.
2. **Actor:** Educator.
3. **Pre:** Unit exists.
4. **Post:** Problem added.
5. **Normal Flow:** Adds Problem, selects "Multi-Select", enters options and marks multiple correct answers.
6. **Alternative Flows:** None.
7. **Business Rules:** Multiple correct answers; partial credit possible.
8. **Input/Output:** Input: question, options, correct options. Output: problem stored.
9. **Acceptance Criteria:**

Given I am an educator
When I create a multi-select problem
Then the problem allows selection of multiple options and grades accordingly

Requirement UFR-OE-005-03: Open Response Assessments (ORA)

1. **Title:** Create open response assessment.
2. **UML Traceability:** Activity: "ORA Flow".
3. **Actor:** Educator.
4. **Pre:** Unit exists.
5. **Post:** ORA created.
6. **Normal Flow:**
 - (a) Adds Advanced component → "Open Response Assessment".
 - (b) Defines prompt and rubric.

- (c) Configures assessment steps (self, peer, staff).
- (d) Sets deadlines.
- (e) Saves.

7. Alternative Flows: None.

8. Business Rules: Rubric criteria defined; deadlines must be in order.

9. Input/Output: Input: prompt, rubric, steps, deadlines. Output: ORA stored.

10. Acceptance Criteria:

```
Given I am an educator
When I create an ORA with peer assessment
Then learners can submit responses and review peers
```

Requirement UFR-OE-005-04: Single Select Problem

- 1. Title:** Create single select (radio button) problem.
- 2. Actor:** Educator.
- 3. Pre:** Unit exists.
- 4. Post:** Problem added.
- 5. Normal Flow:** Adds Problem, selects “Single Select”, enters options, marks correct answer.
- 6. Alternative Flows:** None.
- 7. Business Rules:** One correct answer.
- 8. Input/Output:** Input: question, options, correct answer. Output: problem stored.
- 9. Acceptance Criteria:**

```
Given I am an educator
When I create a single select problem
Then the problem displays radio buttons and grades accordingly
```

Requirement UFR-OE-005-05: Text Input Problem

- 1. Title:** Create text input problem.
- 2. Actor:** Educator.
- 3. Pre:** Unit exists.
- 4. Post:** Problem added.

5. **Normal Flow:** Adds Problem, selects “Text Input”, enters question and expected answer(s).
6. **Alternative Flows:** None.
7. **Business Rules:** Supports multiple accepted answers (case-insensitive).
8. **Input/Output:** Input: question, expected answers. Output: problem stored.
9. **Acceptance Criteria:**

```
Given I am an educator
When I create a text input problem
Then the learner can type an answer and receive immediate feedback
```

0.3.6 Data & Analytics Module

Requirement UFR-OE-006-01: View Course Enrollments

1. **Title:** View course enrollment data.
2. **UML Traceability:** Use Case: “View Analytics”.
3. **Actor:** Educator.
4. **Pre:** Educator has instructor access.
5. **Post:** Enrollment report displayed.
6. **Normal Flow:** Accesses Instructor Dashboard → Data Downloads → Enrollment Data; views totals, trends, demographics.
7. **Alternative Flows:** None.
8. **Business Rules:** Data is real-time.
9. **Input/Output:** Output: CSV with enrollment details.
10. **Acceptance Criteria:**

```
Given I am an educator
When I download enrollment data
Then I receive a CSV with all active enrollments
```

Requirement UFR-OE-006-02: View Learner Data

1. **Title:** View learner data and activity.
2. **Actor:** Educator.
3. **Pre:** Educator has instructor access.
4. **Post:** Learner information displayed.
5. **Normal Flow:** Instructor Dashboard → Student Admin; searches learner; views progress, engagement, activity history.
6. **Alternative Flows:** None.
7. **Business Rules:** Data is anonymized if required.
8. **Input/Output:** Input: learner identifier. Output: learner profile with activity.
9. **Acceptance Criteria:**

```
Given I am an educator
When I search for a learner
Then I see their progress and activity logs
```

Requirement UFR-OE-006-03: View Learner Grades

1. **Title:** View and manage learner grades.
2. **UML Traceability:** Sequence: “Grade View”.
3. **Actor:** Educator.
4. **Pre:** Educator has instructor access.
5. **Post:** Gradebook displayed.
6. **Normal Flow:** Accesses Gradebook; views grades per assignment; adjusts manually if needed; exports CSV.
7. **Alternative Flows:** None.
8. **Business Rules:** Grade changes are logged.
9. **Input/Output:** Output: gradebook; input: grade adjustments.
10. **Acceptance Criteria:**

```
Given I am an educator
When I adjust a grade in the gradebook
Then the grade is updated and an audit log is created
```

Requirement UFR-OE-006-04: View Answer Data

1. **Title:** View answer distribution data.
2. **Actor:** Educator.
3. **Pre:** Educator has instructor access.
4. **Post:** Distribution displayed.
5. **Normal Flow:** Data Downloads → Answer Distribution; sees % correct per problem.
6. **Alternative Flows:** None.
7. **Business Rules:** Data is aggregated.
8. **Input/Output:** Output: distribution report.
9. **Acceptance Criteria:**

```
Given I am an educator
When I view answer distribution
Then I see the percentage of correct answers for each problem
```

Requirement UFR-OE-006-05: View Certificate Data

1. **Title:** View certificate issuance data.
2. **Actor:** Educator.
3. **Pre:** Educator has instructor access.
4. **Post:** Certificate data displayed.
5. **Normal Flow:** Data Downloads → Certificate Data; views list of issued certificates and download stats.
6. **Alternative Flows:** None.
7. **Business Rules:** Data includes learner names, certificate URLs, and generation dates.
8. **Input/Output:** Output: certificate report.
9. **Acceptance Criteria:**

```
Given I am an educator
When I download certificate data
Then I get a list of all issued certificates
```

0.3.7 Accessibility Module

Requirement UFR-OE-007-01: Open edX Accessibility Guidelines

1. **Title:** Ensure accessibility compliance.
2. **Actor:** Educator.
3. **Pre:** Course content being developed.
4. **Post:** Content meets WCAG 2.1 AA.
5. **Normal Flow:** Educator follows guidelines; uses alt text, transcripts, proper structure.
6. **Alternative Flows:** None.
7. **Business Rules:** Compliance is checked during audit.
8. **Input/Output:** None.
9. **Acceptance Criteria:** All content passes accessibility checks.

Requirement UFR-OE-007-02: Mobile Experience Design

1. **Title:** Design for mobile experience.
2. **Actor:** Educator.
3. **Pre:** Course content created.
4. **Post:** Content is responsive and functional on mobile.
5. **Normal Flow:** Educator tests content on mobile devices; adjusts interactions.
6. **Alternative Flows:** None.
7. **Business Rules:** Must work on iOS and Android.
8. **Input/Output:** None.
9. **Acceptance Criteria:** Course renders correctly on mobile browsers.

Requirement UFR-OE-007-03: Accessibility Best Practices Checklist

1. **Title:** Follow accessibility best practices.
2. **Actor:** Educator.
3. **Pre:** Content created.
4. **Post:** Checklist items satisfied.
5. **Normal Flow:** Educator reviews checklist (alt text, headings, contrast, keyboard navigation, descriptive links).
6. **Alternative Flows:** None.

7. **Business Rules:** All items must be satisfied.
8. **Input/Output:** None.
9. **Acceptance Criteria:** Checklist is completed and verified.

Requirement UFR-OE-007-04: Manage Video Transcripts

1. **Title:** Upload and manage video transcripts.
2. **Actor:** Educator.
3. **Pre:** Video component exists.
4. **Post:** Transcript attached.
5. **Normal Flow:** Uploads .srt/.vtt files for video components; transcripts appear as captions.
6. **Alternative Flows:** None.
7. **Business Rules:** Transcript must be in supported format.
8. **Input/Output:** Input: transcript file. Output: captions enabled.
9. **Acceptance Criteria:**

Given I am an educator
When I upload a transcript for a video
Then the video player shows closed captions

0.3.8 Import/Export / OLX Module

Requirement UFR-OE-008-01: Open Learning XML (OLX) Support

1. **Title:** Support for Open Learning XML.
2. **Actor:** Educator.
3. **Pre:** Course exists.
4. **Post:** Course can be exported/imported.
5. **Normal Flow:** Educator uses export/import features; system generates or consumes OLX archives.
6. **Alternative Flows:** None.
7. **Business Rules:** OLX must be valid.
8. **Input/Output:** Input: OLX archive; output: exported archive.
9. **Acceptance Criteria:**

```
Given I am an educator
When I export a course
Then I receive a .tar.gz file containing OLV files
When I import the same file
Then the course structure is recreated exactly
```

Requirement UFR-OE-008-02: Getting Started with OLV

1. **Title:** Access OLV documentation and tools.
2. **Actor:** Educator.
3. **Pre:** None.
4. **Post:** Educator can learn OLV.
5. **Normal Flow:** System provides links to documentation and command-line tools.
6. **Alternative Flows:** None.
7. **Business Rules:** Documentation must be current.
8. **Input/Output:** None.
9. **Acceptance Criteria:** Documentation is accessible from the platform.

0.4 Functional Requirements – Learner Features

0.4.1 Dashboard & Profile Module

Requirement UFR-OE-101-01: Explore Dashboard and Profile

1. **Title:** Access and explore learner dashboard.
2. **UML Traceability:** Use Case: “View Dashboard”.
3. **Actor:** Learner.
4. **Pre:** Learner logged in.
5. **Post:** Dashboard displays enrolled courses, progress, announcements.
6. **Normal Flow:**
 - (a) Learner logs in.
 - (b) Dashboard shows course cards with progress bars, recent activity, and site announcements.
7. **Alternative Flows:** None.
8. **Business Rules:** Dashboard is personalized.
9. **Input/Output:** Output: dashboard view.

10. Acceptance Criteria:

Given I am a learner
When I log in
Then I see my enrolled courses and progress

Requirement UFR-OE-101-02: Access Courses from Dashboard

1. **Title:** Access enrolled courses from dashboard.
2. **Actor:** Learner.
3. **Pre:** Learner logged in.
4. **Post:** Course opens.
5. **Normal Flow:** Clicks on course card; enters the course.
6. **Alternative Flows:** None.
7. **Business Rules:** None.
8. **Input/Output:** Input: click on card. Output: course home page.
9. **Acceptance Criteria:**

Given I am a learner
When I click on a course card
Then I am taken to the course home page

Requirement UFR-OE-101-03: Share Courses on Social Media

1. **Title:** Share course links on social media.
2. **Actor:** Learner.
3. **Pre:** Course page open.
4. **Post:** Link shared.
5. **Normal Flow:** Clicks “Share” button; selects platform; link posted.
6. **Alternative Flows:** None.
7. **Business Rules:** Must use standard sharing APIs.
8. **Input/Output:** Input: share action. Output: social media post.
9. **Acceptance Criteria:**

Given I am a learner
When I share a course on Facebook
Then the course link appears on my timeline

Requirement UFR-OE-101-04: Update Profile Image

1. **Title:** Upload or update profile image.
2. **Actor:** Learner.
3. **Pre:** Learner logged in.
4. **Post:** Image updated.
5. **Normal Flow:** Clicks on profile, selects image file; system resizes and stores.
6. **Alternative Flows:** None.
7. **Business Rules:** Image must be under 5MB; supported formats: PNG, JPG.
8. **Input/Output:** Input: image file. Output: updated profile.
9. **Acceptance Criteria:**

```
Given I am a learner
When I upload a new profile image
Then my profile displays the new image
```

Requirement UFR-OE-101-05: Update Profile Details

1. **Title:** Update profile information.
2. **Actor:** Learner.
3. **Pre:** Learner logged in.
4. **Post:** Profile updated.
5. **Normal Flow:** Edits fields (name, email, location, bio, language) and saves.
6. **Alternative Flows:** None.
7. **Business Rules:** Email must be unique.
8. **Input/Output:** Input: updated fields. Output: saved profile.
9. **Acceptance Criteria:**

```
Given I am a learner
When I change my bio and save
Then my profile shows the new bio
```

Requirement UFR-OE-101-06: Add Social Media Links

1. **Title:** Add personal social media links.
2. **Actor:** Learner.
3. **Pre:** Profile edit page open.
4. **Post:** Links added.
5. **Normal Flow:** Adds URLs for Twitter, LinkedIn, GitHub, etc.
6. **Alternative Flows:** None.
7. **Business Rules:** URLs must be valid.
8. **Input/Output:** Input: URLs. Output: icons on profile.
9. **Acceptance Criteria:**

```
Given I am a learner
When I add my Twitter URL
Then the Twitter icon appears on my profile
```

Requirement UFR-OE-101-07: Toggle Profile Visibility

1. **Title:** Control profile visibility.
2. **Actor:** Learner.
3. **Pre:** Profile edit page open.
4. **Post:** Visibility updated.
5. **Normal Flow:** Toggles visibility of profile sections (public/private).
6. **Alternative Flows:** None.
7. **Business Rules:** Default is private.
8. **Input/Output:** Input: toggle setting. Output: updated visibility.
9. **Acceptance Criteria:**

```
Given I am a learner
When I set my profile to public
Then other users can see my profile information
```

Requirement UFR-OE-101-08: View Another Learner's Profile

1. **Title:** View other learners' profiles.
2. **Actor:** Learner.
3. **Pre:** Another learner's username is visible.
4. **Post:** Profile displayed.
5. **Normal Flow:** Clicks on username in discussions or roster; profile opens (public info only).
6. **Alternative Flows:** None.
7. **Business Rules:** Only public information is shown.
8. **Input/Output:** Input: username. Output: profile view.
9. **Acceptance Criteria:**

```
Given I am a learner
When I click on another learner's username
Then I see their public profile information
```

Requirement UFR-OE-101-09: Update Course-Specific Settings

1. **Title:** Update course-specific settings.
2. **Actor:** Learner.
3. **Pre:** Enrolled in a course.
4. **Post:** Settings updated.
5. **Normal Flow:** In course, clicks avatar → Course Settings; adjusts email notifications, language.
6. **Alternative Flows:** None.
7. **Business Rules:** Settings are per course.
8. **Input/Output:** Input: settings changes. Output: saved.
9. **Acceptance Criteria:**

```
Given I am a learner enrolled in a course
When I disable email notifications
Then I no longer receive course emails
```

0.4.2 Course Navigation Module

Requirement UFR-OE-102-01: View Course Sections from Navigation Sidebar

1. **Title:** Navigate course using sidebar.
2. **Actor:** Learner.
3. **Pre:** Enrolled in a course.
4. **Post:** Navigation displays sections.
5. **Normal Flow:** Sidebar displays all sections/subsections; click to navigate.
6. **Alternative Flows:** None.
7. **Business Rules:** Only published content is visible.
8. **Input/Output:** Input: click on item. Output: content displayed.
9. **Acceptance Criteria:**

```
Given I am a learner in a course
When I click on a section in the sidebar
Then I am taken to that section's content
```

0.4.3 Enrollment Module

Requirement UFR-OE-103-01: Enroll in a Course

1. **Title:** Enroll in a course.
2. **UML Traceability:** Sequence: “Enrollment Flow”.
3. **Actor:** Learner.
4. **Pre:** Course open for enrollment.
5. **Post:** Learner enrolled.
6. **Normal Flow:**
 - (a) Browses course catalog.
 - (b) Clicks “Enroll Now”.
 - (c) Selects enrollment track (Audit/Verified).
 - (d) Confirms enrollment.
7. **Alternative Flows:** None.
8. **Business Rules:** Enrollment may require payment for verified track.
9. **Input/Output:** Input: course selection, track. Output: enrollment confirmation.

10. Acceptance Criteria:

Given I am a learner
When I enroll in a course
Then the course appears in my dashboard

Requirement UFR-OE-103-02: Open and Start a Course

1. **Title:** Open and start a course.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Course opens to first content.
5. **Normal Flow:** Clicks “Start Course” or “Resume Course” from dashboard; goes to last visited content.
6. **Alternative Flows:** None.
7. **Business Rules:** Resume picks last visited unit.
8. **Input/Output:** Input: click start/resume. Output: course content.
9. **Acceptance Criteria:**

Given I am a learner enrolled in a course
When I click "Resume Course"
Then I am taken to the last content I viewed

0.4.4 Course Prerequisites Module

Requirement UFR-OE-104-01: Overview of Prerequisites

1. **Title:** View course prerequisites.
2. **Actor:** Learner.
3. **Pre:** Course info page.
4. **Post:** Prerequisites listed.
5. **Normal Flow:** Prerequisites displayed on course info page; learner can check completion status.
6. **Alternative Flows:** None.
7. **Business Rules:** Prerequisites are defined by educator.
8. **Input/Output:** Output: prerequisite list.

9. Acceptance Criteria:

Given I am a learner
When I view a course with prerequisites
Then I see the required courses and my completion status

Requirement UFR-OE-104-02: Complete Prerequisite Course

1. **Title:** Complete prerequisite course.
2. **Actor:** Learner.
3. **Pre:** Prerequisite course defined.
4. **Post:** Prerequisite marked complete.
5. **Normal Flow:** Must complete prerequisite course before enrolling in advanced course.
6. **Alternative Flows:** None.
7. **Business Rules:** Completion is based on passing grade.
8. **Input/Output:** None.
9. **Acceptance Criteria:**

Given I am a learner
When I complete a prerequisite course with a passing grade
Then the advanced course becomes available

Requirement UFR-OE-104-03: Pass Entrance Exam

1. **Title:** Pass entrance exam.
2. **Actor:** Learner.
3. **Pre:** Course requires entrance exam.
4. **Post:** Exam passed.
5. **Normal Flow:** Takes and passes an entrance exam to gain access.
6. **Alternative Flows:** If fail, retake allowed after cooldown.
7. **Business Rules:** Minimum score set by educator.
8. **Input/Output:** Input: exam answers. Output: pass/fail.
9. **Acceptance Criteria:**

Given I am a learner
When I pass the entrance exam
Then I can access the course content

0.4.5 Course Content Access Module

Requirement UFR-OE-105-01: Troubleshoot Accessing Course Content

1. **Title:** Troubleshoot content access issues.
2. **Actor:** Learner.
3. **Pre:** Content not accessible.
4. **Post:** Issue resolved or guidance given.
5. **Normal Flow:** System provides error messages and support links; contact course staff.
6. **Alternative Flows:** None.
7. **Business Rules:** Error messages are user-friendly.
8. **Input/Output:** Output: error message.
9. **Acceptance Criteria:**

```
Given I am a learner
When I try to access locked content
Then I see a message explaining why and how to gain access
```

Requirement UFR-OE-105-02: Access an Archived Course

1. **Title:** Access archived course.
2. **Actor:** Learner.
3. **Pre:** Course archived.
4. **Post:** Read-only access.
5. **Normal Flow:** Views read-only archived course content.
6. **Alternative Flows:** None.
7. **Business Rules:** No submissions allowed.
8. **Input/Output:** Input: click on archived course. Output: read-only view.
9. **Acceptance Criteria:**

```
Given I am a learner
When I access an archived course
Then I can view content but not submit assignments
```

Requirement UFR-OE-105-03: View Final Course Grade

1. **Title:** View final course grade.
2. **Actor:** Learner.
3. **Pre:** Course completed.
4. **Post:** Grade displayed.
5. **Normal Flow:** Goes to course dashboard; clicks Grades; final grade displayed.
6. **Alternative Flows:** None.
7. **Business Rules:** Grade calculated from all assessments.
8. **Input/Output:** Output: final grade.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view my course grades
Then I see my final grade and passing status
```

Requirement UFR-OE-105-04: View Certificate Status

1. **Title:** View certificate status.
2. **Actor:** Learner.
3. **Pre:** Course completed.
4. **Post:** Status displayed.
5. **Normal Flow:** In profile → Certificates, sees status (Eligible, Available, Not Eligible).
6. **Alternative Flows:** None.
7. **Business Rules:** Certificate available if passing grade.
8. **Input/Output:** Output: certificate status.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view my certificate status
Then I see if I am eligible and can download it
```


0.4.6 Course Progress Module

Requirement UFR-OE-106-01: Find Course Start Date

1. **Title:** View course start date.
2. **Actor:** Learner.
3. **Pre:** Course info page.
4. **Post:** Start date displayed.
5. **Normal Flow:** Start date displayed on course info and dashboard.
6. **Alternative Flows:** None.
7. **Business Rules:** Start date set by educator.
8. **Input/Output:** Output: start date.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view a course
Then I see the start date
```

Requirement UFR-OE-106-02: Start or Resume a Course

1. **Title:** Start or resume a course.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Course content opened.
5. **Normal Flow:** Clicks “Start” or “Resume” from dashboard; goes to last visited content.
6. **Alternative Flows:** None.
7. **Business Rules:** Resume picks last unit.
8. **Input/Output:** Input: click. Output: content.
9. **Acceptance Criteria:**

```
Given I am a learner
When I click "Resume"
Then I am taken to the last unit I viewed
```

Requirement UFR-OE-106-03: Progress Indicators

1. **Title:** View progress indicators.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Progress shown.
5. **Normal Flow:** Visual indicators show completion percentage for sections, subsections, and overall.
6. **Alternative Flows:** None.
7. **Business Rules:** Progress is calculated automatically.
8. **Input/Output:** Output: progress bars.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view a course section
Then I see a progress bar indicating completion
```

Requirement UFR-OE-106-04: The Progress Page

1. **Title:** Access the Progress page.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Detailed progress displayed.
5. **Normal Flow:** Navigates to Progress page to view detailed completion status.
6. **Alternative Flows:** None.
7. **Business Rules:** Shows per-assignment scores.
8. **Input/Output:** Output: progress detail.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view the Progress page
Then I see scores for each assignment and overall completion
```

0.4.7 Certificates & Badges Module

Requirement UFR-OE-107-01: Accessing a Certificate

1. **Title:** Access earned certificate.
2. **Actor:** Learner.
3. **Pre:** Certificate available.
4. **Post:** Certificate downloaded.
5. **Normal Flow:** Goes to profile → Certificates → download.
6. **Alternative Flows:** None.
7. **Business Rules:** Certificate is PDF with digital signature.
8. **Input/Output:** Input: click download. Output: PDF file.
9. **Acceptance Criteria:**

```
Given I am a learner with an available certificate
When I click download
Then a PDF certificate is generated and downloaded
```

Requirement UFR-OE-107-02: Viewing Earned Badges in Profile

1. **Title:** View earned badges in profile.
2. **Actor:** Learner.
3. **Pre:** Badges earned.
4. **Post:** Badges displayed.
5. **Normal Flow:** Badges appear in profile; click to view details.
6. **Alternative Flows:** None.
7. **Business Rules:** Badges are awarded based on achievements.
8. **Input/Output:** Output: badge list.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view my profile
Then I see all earned badges
```

Requirement UFR-OE-107-03: Sharing Badges on Mozilla Backpack

1. **Title:** Share badges to Mozilla Backpack.
2. **Actor:** Learner.
3. **Pre:** Badge earned.
4. **Post:** Badge shared.
5. **Normal Flow:** Clicks “Share to Backpack” and authorises.
6. **Alternative Flows:** None.
7. **Business Rules:** Requires Mozilla Backpack account.
8. **Input/Output:** Input: share action. Output: badge added to Backpack.
9. **Acceptance Criteria:**

Given I am a learner
When I share a badge to Mozilla Backpack
Then the badge appears in my Backpack

0.4.8 Video Player Module

Requirement UFR-OE-108-01: Watching Videos on the Open edX Video Player

1. **Title:** Use the video player.
2. **Actor:** Learner.
3. **Pre:** Video component.
4. **Post:** Video plays.
5. **Normal Flow:** Player controls: play, pause, volume, speed, full-screen, CC/subtitles.
6. **Alternative Flows:** None.
7. **Business Rules:** Video must be compatible.
8. **Input/Output:** Input: user interactions. Output: video playback.
9. **Acceptance Criteria:**

Given I am a learner
When I play a video
Then I can control playback and see captions if available

0.4.9 Discussions Module

Requirement UFR-OE-109-01: Getting Started with Discussions

1. **Title:** Access course discussions.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Discussions view.
5. **Normal Flow:** Clicks “Discussions” tab; sees threads and topics.
6. **Alternative Flows:** None.
7. **Business Rules:** Discussions enabled by educator.
8. **Input/Output:** Output: forum view.
9. **Acceptance Criteria:**

```
Given I am a learner
When I click "Discussions"
Then I see all threads and can create new posts
```

Requirement UFR-OE-109-02: Finding and Following Conversations

1. **Title:** Find and follow conversations.
2. **Actor:** Learner.
3. **Pre:** Discussions open.
4. **Post:** Followed thread.
5. **Normal Flow:** Search, filter, follow threads, get notifications.
6. **Alternative Flows:** None.
7. **Business Rules:** Following sends email notifications.
8. **Input/Output:** Input: follow action. Output: notification preference set.
9. **Acceptance Criteria:**

```
Given I am a learner
When I follow a discussion thread
Then I receive email notifications for new replies
```

Requirement UFR-OE-109-03: Taking Part in Discussions

1. **Title:** Participate in discussions.
2. **Actor:** Learner.
3. **Pre:** Discussions open.
4. **Post:** Post created.
5. **Normal Flow:** Creates new threads, replies to existing ones.
6. **Alternative Flows:** None.
7. **Business Rules:** Posts are moderated.
8. **Input/Output:** Input: post content. Output: visible post.
9. **Acceptance Criteria:**

```
Given I am a learner
When I reply to a thread
Then my reply appears in the discussion
```

Requirement UFR-OE-109-04: Staying Updated with Notifications

1. **Title:** Receive discussion notifications.
2. **Actor:** Learner.
3. **Pre:** Following threads.
4. **Post:** Notification received.
5. **Normal Flow:** System sends email notifications for followed threads.
6. **Alternative Flows:** None.
7. **Business Rules:** Notifications respect user preferences.
8. **Input/Output:** Output: email.
9. **Acceptance Criteria:**

```
Given I am a learner following a thread
When someone replies
Then I receive an email notification
```

0.4.10 Assignments & Assessments Module

Requirement UFR-OE-110-01: Completing Mathematical and Scientific Assignments

1. **Title:** Complete mathematical assignments.
2. **Actor:** Learner.
3. **Pre:** Assignment open.
4. **Post:** Answer submitted.
5. **Normal Flow:** Uses LaTeX for math expressions.
6. **Alternative Flows:** None.
7. **Business Rules:** LaTeX must be valid.
8. **Input/Output:** Input: answer with LaTeX. Output: graded answer.
9. **Acceptance Criteria:**

```
Given I am a learner
When I submit an answer with LaTeX
Then the math renders correctly and is graded
```

Requirement UFR-OE-110-02: Taking a Timed Exam

1. **Title:** Take timed exam.
2. **Actor:** Learner.
3. **Pre:** Exam open.
4. **Post:** Exam submitted.
5. **Normal Flow:** Starts exam; timer counts down; auto-submit at expiration.
6. **Alternative Flows:** None.
7. **Business Rules:** Time limit set by educator.
8. **Input/Output:** Input: answers. Output: grade.
9. **Acceptance Criteria:**

```
Given I am a learner
When I start a timed exam
Then I must submit before time expires or it auto-submits
```

Requirement UFR-OE-110-03: Proctored Exams

1. **Title:** Take proctored exam.
2. **Actor:** Learner.
3. **Pre:** Proctored exam assigned.
4. **Post:** Exam taken with monitoring.
5. **Normal Flow:** Installs proctoring extension; verifies identity; takes exam; monitoring active.
6. **Alternative Flows:** None.
7. **Business Rules:** Proctoring software must be installed.
8. **Input/Output:** Input: identity verification. Output: exam session.
9. **Acceptance Criteria:**

```
Given I am a learner
When I take a proctored exam
Then my session is monitored and recorded
```

Requirement UFR-OE-110-04: Explaining Multiple Choice Answers

1. **Title:** Provide explanation for multiple choice answers.
2. **Actor:** Learner.
3. **Pre:** Multiple choice question.
4. **Post:** Explanation saved.
5. **Normal Flow:** Selects answer and optionally enters explanation.
6. **Alternative Flows:** None.
7. **Business Rules:** Explanation optional.
8. **Input/Output:** Input: explanation. Output: stored with answer.
9. **Acceptance Criteria:**

```
Given I am a learner
When I select an answer and add an explanation
Then the explanation is stored for review
```


Requirement UFR-OE-110-05: Completing Essay Assignments

1. **Title:** Complete essay assignment.
2. **Actor:** Learner.
3. **Pre:** Assignment open.
4. **Post:** Essay submitted.
5. **Normal Flow:** Writes essay in text editor (supports formatting).
6. **Alternative Flows:** None.
7. **Business Rules:** Minimum length may be set.
8. **Input/Output:** Input: essay text. Output: submission confirmation.
9. **Acceptance Criteria:**

```
Given I am a learner
When I submit an essay
Then it is saved and available for grading
```

Requirement UFR-OE-110-06: Open Response Assessment Steps

1. **Title:** Follow ORA workflow.
2. **Actor:** Learner.
3. **Pre:** ORA assigned.
4. **Post:** Assessment completed.
5. **Normal Flow:** Steps: submit response, self-assessment, peer assessment, receive score.
6. **Alternative Flows:** None.
7. **Business Rules:** Order of steps fixed.
8. **Input/Output:** Input: response, assessments. Output: final score.
9. **Acceptance Criteria:**

```
Given I am a learner
When I complete all ORA steps
Then I receive a final score
```

Requirement UFR-OE-110-07: How Grading Is Done In Open Response Assessments

1. **Title:** Understand ORA grading.
2. **Actor:** Learner.
3. **Pre:** ORA available.
4. **Post:** Grading explanation shown.
5. **Normal Flow:** System displays grading rubric and peer review scores.
6. **Alternative Flows:** None.
7. **Business Rules:** Rubric defined by educator.
8. **Input/Output:** Output: grading description.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view the ORA grading criteria
Then I see the rubric and how scores are calculated
```

Requirement UFR-OE-110-08: Completing an Open Response Assessment

1. **Title:** Complete ORA.
2. **Actor:** Learner.
3. **Pre:** ORA open.
4. **Post:** ORA completed.
5. **Normal Flow:** Submits response, performs self/peer assessments.
6. **Alternative Flows:** None.
7. **Business Rules:** Peer assessment requires reviewing others.
8. **Input/Output:** Input: response, peer reviews. Output: final score.
9. **Acceptance Criteria:**

```
Given I am a learner
When I complete an ORA
Then I see my final grade and feedback
```

Requirement UFR-OE-110-09: Receive Score and Provide Feedback

1. **Title:** Receive ORA score and provide feedback.
2. **Actor:** Learner.
3. **Pre:** Assessment done.
4. **Post:** Score displayed.
5. **Normal Flow:** Views score and provides feedback on assessment experience.
6. **Alternative Flows:** None.
7. **Business Rules:** Feedback optional.
8. **Input/Output:** Input: feedback. Output: stored.
9. **Acceptance Criteria:**

Given I am a learner
When I receive my ORA score
Then I can optionally submit feedback on the process

Requirement UFR-OE-110-10: How Peer Assessment Scores Are Calculated

1. **Title:** Understand peer assessment scoring.
2. **Actor:** Learner.
3. **Pre:** ORA with peer assessment.
4. **Post:** Scoring explained.
5. **Normal Flow:** System explains scoring algorithm (average, median, adjusted).
6. **Alternative Flows:** None.
7. **Business Rules:** Outliers may be removed.
8. **Input/Output:** Output: scoring description.
9. **Acceptance Criteria:**

Given I am a learner
When I view how peer scores are calculated
Then I understand the algorithm used

Requirement UFR-OE-110-11: Canceled Responses

1. **Title:** Handle canceled responses.
2. **Actor:** Learner.
3. **Pre:** Response submitted.
4. **Post:** Cancellation confirmed.
5. **Normal Flow:** Can cancel submission before assessment.
6. **Alternative Flows:** None.
7. **Business Rules:** Cancellation allowed before assessment phase.
8. **Input/Output:** Input: cancel action. Output: submission removed.
9. **Acceptance Criteria:**

```
Given I am a learner
When I cancel my submission before assessment
Then I can resubmit later
```

0.4.11 Team Projects Module

Requirement UFR-OE-111-01: About Teams and Topics

1. **Title:** Understand teams and topics.
2. **Actor:** Learner.
3. **Pre:** Course with team feature.
4. **Post:** Teams explained.
5. **Normal Flow:** System provides team and topic overview.
6. **Alternative Flows:** None.
7. **Business Rules:** Topics defined by educator.
8. **Input/Output:** Output: description.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view the Teams page
Then I see an explanation of teams and topics
```

Requirement UFR-OE-111-02: Search for a Team

1. **Title:** Search for a team.
2. **Actor:** Learner.
3. **Pre:** Teams page.
4. **Post:** Search results.
5. **Normal Flow:** Searches teams by name, topic, or member.
6. **Alternative Flows:** None.
7. **Business Rules:** Search is case-insensitive.
8. **Input/Output:** Input: search term. Output: list of teams.
9. **Acceptance Criteria:**

```
Given I am a learner
When I search for a team name
Then I see matching teams
```

Requirement UFR-OE-111-03: Join a Team

1. **Title:** Join an existing team.
2. **Actor:** Learner.
3. **Pre:** Team visible.
4. **Post:** Added to team.
5. **Normal Flow:** Clicks “Join Team”; may require approval.
6. **Alternative Flows:** None.
7. **Business Rules:** Team size may be limited.
8. **Input/Output:** Input: join request. Output: membership.
9. **Acceptance Criteria:**

```
Given I am a learner
When I join a team
Then I become a member and can access team discussions
```

Requirement UFR-OE-111-04: Leave a Team

1. **Title:** Leave a team.
2. **Actor:** Learner.
3. **Pre:** Member of team.
4. **Post:** Removed from team.
5. **Normal Flow:** Clicks “Leave Team” and confirms.
6. **Alternative Flows:** None.
7. **Business Rules:** Cannot leave if only member (may dissolve team).
8. **Input/Output:** Input: leave action. Output: removed.
9. **Acceptance Criteria:**

```
Given I am a team member
When I leave the team
Then I am removed and lose access to team resources
```

Requirement UFR-OE-111-05: Create a Team

1. **Title:** Create a new team.
2. **Actor:** Learner.
3. **Pre:** Course allows team creation.
4. **Post:** Team created.
5. **Normal Flow:** Enters name, description, topic, enrollment type; clicks Create.
6. **Alternative Flows:** None.
7. **Business Rules:** Team name unique within course.
8. **Input/Output:** Input: team details. Output: new team.
9. **Acceptance Criteria:**

```
Given I am a learner
When I create a team with a unique name
Then the team appears in the team list
```

Requirement UFR-OE-111-06: Participating in Team Discussions

1. **Title:** Participate in team discussions.
2. **Actor:** Learner.
3. **Pre:** Member of team.
4. **Post:** Discussion posted.
5. **Normal Flow:** Team members discuss within team-specific forum.
6. **Alternative Flows:** None.
7. **Business Rules:** Team discussions are private.
8. **Input/Output:** Input: post. Output: visible to team.
9. **Acceptance Criteria:**

```
Given I am a team member
When I post in team discussion
Then only team members can see my post
```

0.4.12 Search Module

Requirement UFR-OE-112-01: Overview of Search

1. **Title:** Understand search capabilities.
2. **Actor:** Learner.
3. **Pre:** None.
4. **Post:** Search explained.
5. **Normal Flow:** System provides help text about full-text search.
6. **Alternative Flows:** None.
7. **Business Rules:** Search powered by Meilisearch.
8. **Input/Output:** Output: help text.
9. **Acceptance Criteria:**

```
Given I am a learner
When I look for search help
Then I see a description of what is searchable
```

Requirement UFR-OE-112-02: Search a Single Course

1. **Title:** Search within a course.
2. **Actor:** Learner.
3. **Pre:** In a course.
4. **Post:** Results displayed.
5. **Normal Flow:** Enters query; returns relevant content within course.
6. **Alternative Flows:** None.
7. **Business Rules:** Search scope is the current course.
8. **Input/Output:** Input: query. Output: results.
9. **Acceptance Criteria:**

```
Given I am a learner in a course
When I search for a term
Then I see results only from that course
```

Requirement UFR-OE-112-03: Search All Your Courses

1. **Title:** Search across all courses.
2. **Actor:** Learner.
3. **Pre:** On dashboard.
4. **Post:** Results displayed.
5. **Normal Flow:** Enters query; returns results from all enrolled courses.
6. **Alternative Flows:** None.
7. **Business Rules:** Search scope is all enrolled courses.
8. **Input/Output:** Input: query. Output: results.
9. **Acceptance Criteria:**

```
Given I am a learner on the dashboard
When I search for a term
Then I see results from all my courses
```


0.4.13 Bookmarking Module

Requirement UFR-OE-113-01: Add or Remove a Bookmark

1. **Title:** Bookmark course content.
2. **Actor:** Learner.
3. **Pre:** Viewing content.
4. **Post:** Bookmark toggled.
5. **Normal Flow:** Clicks bookmark icon to save/remove bookmark.
6. **Alternative Flows:** None.
7. **Business Rules:** One bookmark per user per content.
8. **Input/Output:** Input: click icon. Output: bookmark status.
9. **Acceptance Criteria:**

```
Given I am a learner viewing content
When I click the bookmark icon
Then the content is bookmarked and appears in my bookmarks list
```

Requirement UFR-OE-113-02: View Your Bookmarks

1. **Title:** View bookmarked content.
2. **Actor:** Learner.
3. **Pre:** Bookmarks exist.
4. **Post:** List displayed.
5. **Normal Flow:** Accesses bookmarks page to view and navigate saved items.
6. **Alternative Flows:** None.
7. **Business Rules:** Sorted by date added.
8. **Input/Output:** Output: bookmark list.
9. **Acceptance Criteria:**

```
Given I have bookmarks
When I view my bookmarks
Then I see a list of all bookmarked content
```

0.4.14 Notifications & Preferences Module

Requirement UFR-OE-114-01: Where Notifications Appear

1. **Title:** Access notifications.
2. **Actor:** Learner.
3. **Pre:** Logged in.
4. **Post:** Notifications visible.
5. **Normal Flow:** Notifications appear in dashboard, email, and in-app alerts.
6. **Alternative Flows:** None.
7. **Business Rules:** In-app via bell icon.
8. **Input/Output:** Output: notification display.
9. **Acceptance Criteria:**

```
Given I am a learner
When I have unread notifications
Then I see the bell icon with a badge count
```

Requirement UFR-OE-114-02: Activity Types

1. **Title:** Understand notification activity types.
2. **Actor:** Learner.
3. **Pre:** None.
4. **Post:** Types explained.
5. **Normal Flow:** System lists notification types (discussion replies, due dates, updates, feedback).
6. **Alternative Flows:** None.
7. **Business Rules:** Types defined by system.
8. **Input/Output:** Output: list of types.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view notification settings
Then I see all available activity types
```

Requirement UFR-OE-114-03: Managing Preferences

1. **Title:** Manage notification preferences.
2. **Actor:** Learner.
3. **Pre:** Logged in.
4. **Post:** Preferences saved.
5. **Normal Flow:** Configures which notifications to receive and via which channels.
6. **Alternative Flows:** None.
7. **Business Rules:** Preferences stored per user.
8. **Input/Output:** Input: settings. Output: saved.
9. **Acceptance Criteria:**

```
Given I am a learner
When I change notification preferences
Then my preferences are updated
```

Requirement UFR-OE-114-04: Notifications Expiry

1. **Title:** Notification expiry.
2. **Actor:** System.
3. **Pre:** Notification older than 30 days.
4. **Post:** Notification archived.
5. **Normal Flow:** Notifications expire after 30 days.
6. **Alternative Flows:** None.
7. **Business Rules:** Archived notifications can still be viewed.
8. **Input/Output:** None.
9. **Acceptance Criteria:**

```
Given a notification is 30 days old
When the system runs the expiry job
Then the notification is moved to archive
```

0.4.15 Course Wiki Module

Requirement UFR-OE-115-01: Using the Course Wiki

1. **Title:** Use the course wiki.
2. **Actor:** Learner.
3. **Pre:** Course has wiki enabled.
4. **Post:** Wiki displayed.
5. **Normal Flow:** Accesses wiki from course navigation; views and edits (if allowed).
6. **Alternative Flows:** None.
7. **Business Rules:** Edit permissions controlled by educator.
8. **Input/Output:** Input: edit (if allowed). Output: wiki page.
9. **Acceptance Criteria:**

```
Given I am a learner
When I open the course wiki
Then I can view and optionally edit pages
```

Requirement UFR-OE-115-02: View Live Course

1. **Title:** View live course content.
2. **Actor:** Learner.
3. **Pre:** Content published.
4. **Post:** Content visible.
5. **Normal Flow:** Accesses content that is currently active/published.
6. **Alternative Flows:** None.
7. **Business Rules:** Only published and released content is visible.
8. **Input/Output:** Input: navigation. Output: content.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view a course
Then I only see content that has been published and released
```

0.5 Security and Compliance Requirements (Non-Functional)

This section defines mandatory security requirements derived from INSA's CMCSRS, the Computer Crime Proclamation, and the Electronic Signature Proclamation. All requirements are traceable to the *Security Architecture & Threat Model* document.

0.5.1 Access Control (CMCSRS 6.17)

- SEC-NFR-001-1. The system must enforce role-based access control (RBAC) for all user actions.
- SEC-NFR-001-2. All access to administrative functions must be restricted to users with appropriate privileges.
- SEC-NFR-001-3. User accounts must be subject to a formal provisioning and de-provisioning process.
- SEC-NFR-001-4. Default passwords for any system component must be changed upon first use.
- SEC-NFR-001-5. The system must implement multi-factor authentication (MFA) for administrative accounts.

0.5.2 Cryptography (CMCSRS 6.18)

- SEC-NFR-002-1. All data in transit must be encrypted using TLS 1.2 or higher.
- SEC-NFR-002-2. All sensitive data at rest must be encrypted using strong, INSA-approved algorithms.
- SEC-NFR-002-3. Private keys and secrets must never be stored in plaintext.
- SEC-NFR-002-4. Digital signatures must be used for certificate verification as per the Electronic Signature Proclamation.

0.5.3 Network Security (CMCSRS 5.7.c)

- SEC-NFR-003-1. A host-based firewall must be active and configured to allow only necessary ports.
- SEC-NFR-003-2. The internal network must be segmented into separate Docker networks.
- SEC-NFR-003-3. All unnecessary services and ports must be disabled.
- SEC-NFR-003-4. Intrusion detection/prevention systems must be deployed.

0.5.4 Logging and Monitoring (CMCSRS 6.25)

- SEC-NFR-004-1. All logs must be collected centrally and retained for at least one month.
- SEC-NFR-004-2. Security-relevant events must be logged with timestamps and user identifiers.
- SEC-NFR-004-3. Regular vulnerability scanning must be performed weekly.
- SEC-NFR-004-4. Anomaly detection and alerting must be configured.

0.5.5 Incident Response and Business Continuity (CMCSRS 6.26, 6.19)

- SEC-NFR-005-1. An incident response plan must be documented and tested annually.
- SEC-NFR-005-2. A business continuity plan with defined RTO and RPO must be maintained.
- SEC-NFR-005-3. Regular backups must be performed and stored off-site.
- SEC-NFR-005-4. All security incidents must be reported to Ethio-CERT and INSA.

0.5.6 Personnel Security (CMCSRS 5.6, 6.37)

- SEC-NFR-006-1. All personnel with access to sensitive systems must undergo security clearance.
- SEC-NFR-006-2. Mandatory annual security awareness training must be provided.
- SEC-NFR-006-3. Job descriptions must include security responsibilities.

0.5.7 Compliance and Audit (CMCSRS 6.21, 6.22, 6.23)

- SEC-NFR-007-1. The system must undergo an annual security audit by an INSA-certified auditor.
- SEC-NFR-007-2. All information systems must be accredited before deployment.
- SEC-NFR-007-3. A compliance report must be submitted to INSA annually.

0.6 Non-Functional Requirements

0.6.1 Performance

- Support at least 10,000 concurrent learners and 500 concurrent educators.
- Page load times below 2 seconds for 95% of requests.
- API response times below 500ms for 95% of requests.
- Video streaming supports at least 720p with adaptive bitrate.

0.6.2 Security

- All network communication encrypted with TLS 1.2+.
- Passwords hashed using PBKDF2/BCrypt.
- API authentication using JWT with short expiry.
- RBAC enforced.
- Firewall and intrusion detection in place.

0.6.3 Availability

- 99.9% uptime.
- Automatic container restarts.
- Daily off-site backups.

0.6.4 Maintainability

- Containerised microservices architecture.
- Quarterly upstream merges.
- Externalised configuration.

0.6.5 Scalability

- Horizontal scaling with container replicas.
- Database read replicas supported.

0.6.6 Usability

- Responsive interface (modern browsers).
- User-friendly error messages.
- WCAG 2.1 AA accessibility.

0.7 Traceability Matrix

Requirement ID	Test Case ID	Security Reference
UFR-OE-001-01	TC-001	SEC-NFR-001, SEC-NFR-002
UFR-OE-001-03	TC-002	SEC-NFR-001
UFR-OE-003-01	TC-003	SEC-NFR-003, SEC-NFR-004
UFR-OE-003-02	TC-004	SEC-NFR-004
UFR-OE-004-01	TC-005	SEC-NFR-002
UFR-OE-004-03	TC-006	SEC-NFR-002
UFR-OE-005-03	TC-007	SEC-NFR-001, SEC-NFR-002
UFR-OE-006-03	TC-008	SEC-NFR-001, SEC-NFR-004
UFR-OE-103-01	TC-009	SEC-NFR-001
UFR-OE-106-03	TC-010	—
Continued		

Requirement ID	Test Case ID	Security Reference
UFR-OE-107-01	TC-011	SEC-NFR-002
UFR-OE-109-01	TC-012	SEC-NFR-001, SEC-NFR-003
UFR-OE-110-02	TC-013	SEC-NFR-003
UFR-OE-112-02	TC-014	—
UFR-OE-113-01	TC-015	—
UFR-OE-114-03	TC-016	SEC-NFR-006
SEC-NFR-001..007	Security doc	CMCSRS, Proc.958/2016

0.8 Compliance Matrix

The following table maps system requirements to specific INSA regulations.

INSA Regulation	Section	System Compliance Evidence
CMCSRS v2.0	5.3 Leadership	Top leadership allocated 0.5% budget; strategic awareness program
CMCSRS v2.0	5.4 Governance	Security committee established; CISO appointed
CMCSRS v2.0	5.5 Management	Risk-based approach implemented; comprehensive risk profiles
CMCSRS v2.0	5.7 Technology	Firewall, network segmentation, vulnerability scanning
CMCSRS v2.0	6.4 Risk Assessment	SGOC+PESTLE, BMIS, ISO 27001 assessments
CMCSRS v2.0	6.17 Access Control	RBAC enforced; non-root user; secrets rotated
CMCSRS v2.0	6.18 Cryptography	TLS 1.2; AES-256; JWT RSA keys rotated
CMCSRS v2.0	6.25 Monitoring	Trivy daily scans; centralised logging; health checks
CMCSRS v2.0	6.26 Incident Response	Fail2ban; IRP documented; reporting to Ethio-CERT
CMCSRS v2.0	6.28 Backup	Daily automated backups; retention policy
Continued		

INSA Regulation	Section	System Compliance Evidence
Computer Crime Proc.	Art 3-8	Access controls; firewall; logging; incident detection
Electronic Signature Proc.	Art 5-8	Digital signatures; certificate verification
National Cyber Policy	2.2 Legal Framework	Compliance with all applicable laws
National Cyber Policy	2.3 Awareness	Security awareness program established
National Cyber Policy	2.4 Capacity Building	Staff training; career path alignment

0.9 Glossary

CMCSRS Critical Mass Cyber Security Requirement Standard.

CMS Course Management System.

DFD Data Flow Diagram.

INSA Information Network Security Administration.

LMS Learning Management System.

MFE Micro-Frontend.

RBAC Role-Based Access Control.

Tutor Open edX deployment tool.

0.10 References

- *Security Architecture & Threat Model – Moove Education Platform* (Security_OpenEdX.tex)
- CMCSRS v2.0 (2025)
- FDRE National Cybersecurity Policy (2024)
- Computer Crime Proclamation No. 958/2016
- Electronic Signature Proclamation No. 1072/2018
- Cyber Security Audit and Evaluation Guideline v1.0
- Cyber Security Risk Assessment Framework v1.0
- Swagger API Documentation (swagger-docs.json)

- Deployment Report (deployment-report.txt)
- VPS Health Check (vps_health_check.txt)

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